LMS Platform: Technical Architecture

Generated Technical Deep-Dive & Q&A Report

1. Executive Summary

This document provides a detailed breakdown of the technological ecosystem powering the LMS Platform. The system is built using a modern, scalable stack prioritized for performance, security, and premium user experience (UX).

2. Core Technology Stack

Frontend Architecture

- Framework: Next.js 15 (App Router) - SEO optimized & high performance.
- Language: TypeScript - Type-safe development across the codebase.
- Styling: Tailwind CSS 4 - Next-gen utility-first CSS.
- UI Components: Radix UI / Shadcn UI - Accessible and professional.
- Animations: Framer Motion - Smooth transitions and micro-interactions.
- Visuals: Three.js & Recharts - 3D Hero sections and data dashboards.

Backend & Infrastructure

- API Server: Express.js (Node.js) - Decoupled business logic.
- ORM: Prisma - Type-safe database access.
- Database: PostgreSQL - Relational storage for structured data.
- Authentication: Clerk - Managed auth with Svix Webhooks.
- Payments: Razorpay & Stripe - Multi-gateway payment processing.

3. Key Implementation Details

- Course Management: Hierarchical structure (Course > Section > Lesson) with granular progress tracking.
- Enrollment Flow: Unified enrollment model triggered by verified payment webhooks.
- UI Philosophy: Consistent, high-fidelity design with physics-based interactivity.

4. Technical Q&A

Q1: Why use Next.js 15 with a separate Express backend?
A: Better scalability for compute-intensive tasks without affecting frontend latency, and flexibility for future cross-platform apps.

Q2: How is user security handled?
A: Fully delegated to Clerk for secure session management, MFA, and hashing, ensuring no sensitive data is stored on-site.

Q3: How is course progress calculated?
A: By real-time tracking of completed lesson nodes vs total course nodes in the PostgreSQL Progress table.

Q4: What is the advantage of using Prisma?
A: Automatic type generation and schema migrations, reducing database-related bugs significantly.